

1. Authentication Endpoints:

1. Register Institution Admin

Endpoint

POST /api/auth/register-admin

Importance

- Allows an institution to request access to the platform.
- Creates a pending admin account that must be approved by the Super Admin.

Not Protected

2. Login

Endpoint

POST /api/auth/login

Importance

- Authenticates users (Super Admin, Admin, Student).
- Returns JWT/session token for accessing protected APIs.

Not Protected

3. Logout

Endpoint

POST /api/auth/logout

Importance

- Invalidates user session or token.
- Ensures security when users log out.

Protected

4. Student Registration

Endpoint

POST /api/auth/register-student

Importance

- Allows students to create an account under a specific institution.
- Enables them to participate in clubs and events.

Not Protected

2. Super Admin APIs

1. Approve Institution

Endpoint

PUT /api/superadmin/institutions/{id}/approve

Importance

- Allows only verified institutions to use the system.
- Prevents unauthorized organizations from accessing the platform.

Protected

2. Reject Institution

Endpoint

PUT /api/superadmin/institutions/{id}/reject

Importance

- Rejects institutions that do not meet requirements.
- Maintains platform integrity.

Protected

3. View All Institutions

Endpoint

GET /api/superadmin/institutions

Importance

- Allows the Super Admin to monitor all registered institutions.
- Helps track subscriptions and system usage.

Protected

4. Deactivate Institution

Endpoint

PUT /api/superadmin/institutions/{id}/deactivate

Importance

- Used when subscription expires or violations occur.
- Prevents the institution from accessing the system.

Protected

3. Club Management APIs

1. Create Club

Endpoint

POST /api/clubs

Importance

- Allows institution admins to create clubs such as coding club, music club, etc.
- Forms the base structure for organizing events.

Protected

2. Get All Clubs

Endpoint

GET /api/clubs

Importance

- Allows students to explore all clubs available in their institution.
- Encourages student engagement.

Protected

3. Update Club

Endpoint

PUT /api/clubs/{clubId}

Importance

- Allows admins to update club information such as description or coordinator.

Protected

4. Delete Club

Endpoint

DELETE /api/clubs/{clubId}

Importance

- Removes inactive or obsolete clubs.

Protected

4. Club Membership APIs

1. Add Student to Club

Endpoint

POST /api/clubs/{clubId}/members

Importance

- Assigns students to clubs.
- Enables them to participate in club activities.

Protected

2. Remove Student from Club

Endpoint

DELETE /api/clubs/{clubId}/members/{studentId}

Importance

- Allows admins to manage club membership.

Protected

5. Event Management APIs

1. Create Event

Endpoint

POST /api/events

Importance

- Enables admins to organize events under a specific club.
- Core functionality of the system.

Protected

2. Get All Events

Endpoint

GET /api/events

Importance

- Allows students to browse all upcoming events.

Protected

3. Get Event Details

Endpoint

GET /api/events/{eventId}

Importance

- Shows event description, participants, entry fee, and schedule.

Protected (Optional)

4. Update Event

Endpoint

PUT /api/events/{eventId}

Importance

- Allows admins to modify event details before it starts.

Protected

5. Delete Event

Endpoint

DELETE /api/events/{eventId}

Importance

- Removes cancelled events.

Protected

6. Event Registration APIs

1. Register for Event

Endpoint

POST /api/events/{eventId}/register

Importance

- Allows students to participate in events.
- Creates participant records.

Protected

2. View My Registrations

Endpoint

GET /api/students/my-events

Importance

- Students can track events they registered for.

Protected

7. Payment APIs

1. Create Payment Order

Endpoint

POST /api/payments/create-order

Importance

- Generates payment order for event registration.

Protected

2. Verify Payment

Endpoint

POST /api/payments/verify

Importance

- Confirms payment success from gateway.

Protected

8. Event Results APIs

1. Publish Winners

Endpoint

POST /api/events/{eventId}/winners

Importance

- Allows admins to declare winners.
- Triggers points allocation to institution.

Protected

2. View Leaderboard

Endpoint

GET /api/leaderboard

Importance

- Displays ranking of institutions based on points.

Not Protected

DB DESIGN:

1. Tenants (Institutions)

Columns

- tenant_id (**PK, Unique**)
- institution_name
- institution_email
- contact_number
- address
- subscription_start_date
- subscription_expiry_date
- status
- created_at

Description

Stores all institutions using the platform.

Each institution is treated as a **tenant**, and tenant_id uniquely identifies the tenant and is used to isolate their data across the system.

2. Users

Columns

- user_id (**PK**)
- tenant_id (**FK** → **Tenants.tenant_id**)
- name
- email (**Unique**)
- password_hash
- role (SUPER_ADMIN / ADMIN / STUDENT)

- status
- created_at

Description

Stores login and profile information for all users including **Super Admins, Institution Admins, and Students**.

tenant_id links users to their institution except for the Super Admin who manages the platform.

3. Clubs

Columns

- club_id (**PK**)
- tenant_id (**FK** → **Tenants.tenant_id**)
- club_name
- description
- created_by (**FK** → **Users.user_id**)
- created_at

Description

Stores all clubs created within an institution (e.g., Coding Club, Photography Club). Each club belongs to a specific tenant and is usually created by an institution admin.

4. Club Members

Columns

- club_member_id (**PK**)
- club_id (**FK** → **Clubs.club_id**)
- student_id (**FK** → **Users.user_id**)
- joined_at

Description

Stores membership information for students who are part of clubs.

This table maintains a **many-to-many relationship between students and clubs**.

5. Events

Columns

- event_id (**PK**)
- tenant_id (**FK** → **Tenants.tenant_id**)
- club_id (**FK** → **Clubs.club_id**)
- event_name
- description
- start_date
- end_date
- entry_fee
- max_participants
- event_type (INSTITUTION / UNIVERSAL)

- created_by (FK → Users.user_id)
- created_at

Description

Stores all event information organized by clubs.

Events can either be **institution-specific** or **universal**, allowing students from other institutions to participate.

6. Event Registrations

Columns

- registration_id (PK)
- event_id (FK → Events.event_id)
- student_id (FK → Users.user_id)
- registration_date
- payment_status
- status

Description

Stores records of students registering for events.

Ensures that participation details and registration status are tracked for each event.

7. Payments

Columns

- payment_id (PK)
- registration_id (FK → EventRegistrations.registration_id)
- student_id (FK → Users.user_id)
- amount
- payment_method
- payment_status
- transaction_reference
- payment_date

Description

Stores payment details for event registrations where an entry fee is required.

Helps track successful and failed transactions from the payment gateway.

8. Event Winners

Columns

- winner_id (PK)
- event_id (FK → Events.event_id)
- student_id (FK → Users.user_id)
- position
- points_awarded

Description

Stores winners of events and the points awarded to them.
These points contribute to the **overall ranking of institutions**.

9. Event Media

Columns

- media_id (**PK**)
- event_id (**FK** → **Events.event_id**)
- media_url
- media_type
- uploaded_at

Description

Stores images or media files related to events such as event photos or promotional material.

10. Institution Points

Columns

- institution_points_id (**PK**)
- tenant_id (**FK** → **Tenants.tenant_id**)
- total_points
- last_updated

Description

Stores the accumulated points earned by an institution through event wins.
This table helps generate the **institution leaderboard**.